

- 8 (a)** When monochromatic light is shone on a clean metal surface, electrons are emitted from the surface due to the photoelectric effect.

(i) State what is meant by the threshold frequency of the incident light.
below a certain frequency (called the threshold frequency) no electrons emitted

.....
..... [1]

(ii) Explain why the photoelectric effect is not observed below the threshold frequency.

(light travels as photons) energy of a photon depends on frequency

below threshold frequency (photon) does not have enough energy

to liberate an electron / (to allow the electron) to overcome the work function

or minimum frequency of the photon that will give the electrons enough energy to overcome work function

..... [2]

- (b) Monochromatic light of wavelength 5.40×10^{-7} m is incident on a metal surface which has a work function of 1.40×10^{-19} J.

(i) Calculate the threshold frequency of this incident light.

$$\phi = hf_0$$

$$f_0 = \phi / h = 1.40 \times 10^{-19} / 6.63 \times 10^{-34}$$

$$= 2.11 \times 10^{14} \text{ Hz}$$

threshold frequency = Hz [2]

(ii) Calculate the energy of a single photon of this light.

$$E = hf = hc/\lambda$$

$$E = 6.63 \times 10^{-34} \times 3.00 \times 10^8 / 5.40 \times 10^{-7}$$

$$E = 3.68 \times 10^{-19} \text{ J}$$

energy of photon = J [2]

(iii) Calculate the maximum speed of the emitted electron.

$$hf = E_k + \phi$$

$$3.68 \times 10^{-19} = E_{k,\text{max}} + 1.40 \times 10^{-19}$$

$$E_{k,\text{max}} = 2.28 \times 10^{-19} \text{ J}$$

$$E_{k,\text{max}} = \frac{1}{2} mv_{\text{max}}^2$$

$$v_{\text{max}}^2 = 2 \times 2.28 \times 10^{-19} / 9.11 \times 10^{-31}$$

$$v_{\text{max}} = 7.1 \times 10^5 \text{ m s}^{-1}$$

maximum speed = m s⁻¹ [3]

- (iv) Calculate the de Broglie wavelength of the fastest electrons.

$$\lambda = h / p = h / mv$$

$$\lambda = 6.63 \times 10^{-34} (9.11 \times 10^{-31} \times 7.1 \times 10^5)$$

$$\lambda = 1.03 \times 10^{-9} = 1.03 \text{ nm}$$

de Broglie wavelength = nm [2]

- (c) Fig. 8.1 below represents the energy levels of the four lowest states of the hydrogen atom.

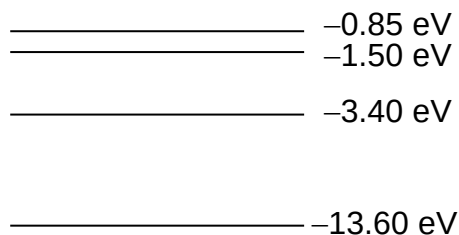


Fig. 8.1

- (i) State what happens in the atom when line spectra are produced.
 electrons are excited or de-excited from one energy level (or orbit) to another

 emitting or absorbing a definite frequency/wavelength/colour or photon
 energy (of electromagnetic radiation)

 [1]

- (ii) State the transition and determine the wavelength of one possible **visible** spectral line detected in the **emission** spectrum of atomic hydrogen, due to transitions between these states.

loss in energy by atom = energy of photon emitted

$$\Delta E = hf = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{\Delta E}$$

Either 4 → 2: 488 nm = 0.488 μm

Or 3 → 2: 654 nm = 0.654 μm

transition = level to

wavelength = μm [3]

- (d) Fig. 8.2 below represents a typical X-ray spectrum.

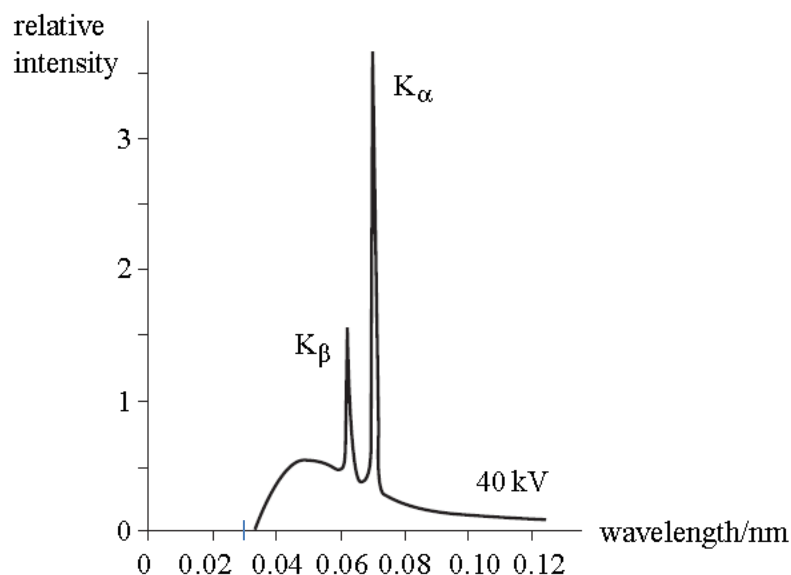


Fig. 8.2

- (i) Distinguish between the mechanisms that produce **characteristic** X-ray spectra in Fig. 8.2 and the line spectra in part (c)(i).

In (c)(i): electrons are excited to higher levels by incoming electrons and subsequently decay to lower levels, emitting photons

.....
in Fig. 8.2: inner electrons ejected by incoming electrons and (outer) electrons fall to unoccupied level

[2]

- (ii) Estimate the maximum velocity of the electrons
loss in $E_K =$ X-ray photon emitted

$$\frac{1}{2}mv^2 = hf = h\frac{c}{\lambda}$$

$$v_{\max} = \sqrt{\frac{2hc}{m\lambda_{\min}}} = \sqrt{\frac{2(6.63 \times 10^{-34})(3.00 \times 10^8)}{(9.11 \times 10^{-31})(\approx 0.034 \times 10^{-9})}}$$

$$\approx 1 \times 10^8 \text{ m s}^{-1}$$

maximum velocity $\approx$ m s⁻¹ [2]

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